

AI Engineering Assessment

Overview

Build a working demo that answers a business question about SEC financial filings using a single LLM API call with retrieval-augmented generation and present it in a simulated client meeting.

Timebox: ~4 hours

The Task

Build a retrieval index over the provided corpus and a prompt template that injects retrieved context into a single LLM call. The system should accept a natural-language business question as input and return a well-structured answer grounded in the filing data.

The final answer must be produced in **one API request**. Your indexing and retrieval pipeline can run beforehand, but the answer itself must come from a single LLM call. Beyond that constraint, the design is yours.

Document your assumptions and the decisions you make along the way.

Dataset

The corpus is a collection of SEC EDGAR filings — annual reports (10-K) and quarterly reports (10-Q) from major US public companies spanning 2023–2025.

Download the corpus here:

[edgar_corpus.zip](#)

The zip contains `.txt` filings and a `manifest.json`.

Business Question

During the demo, we will enter a business question into your system's input field. Examples of what we might ask:

- *"What are the primary risk factors facing Apple, Tesla, and JPMorgan, and how do they compare?"*
- *"How has NVIDIA's revenue and growth outlook changed over the last two years?"*
- *"What regulatory risks do the major pharmaceutical companies face, and how are they addressing them?"*

Deliverables

Submit a repo or notebook containing:

- A README with setup and run instructions
- Your indexing and retrieval code
- A log of your prompt iterations (what changed, why)
- Your final prompt template
- A front-end
- An example request ready to execute
- Notes on how you evaluated quality

Demo Format

You will set up your environment, talk your interview panel through it as if this were a real client meeting, and run a question through your system. In addition to the repo, we want to see a front-end to the system.

Be prepared to walk through your design decisions.

Questions

If anything is unclear, ask. We'd rather answer a good question now than review a set of wrong assumptions later.